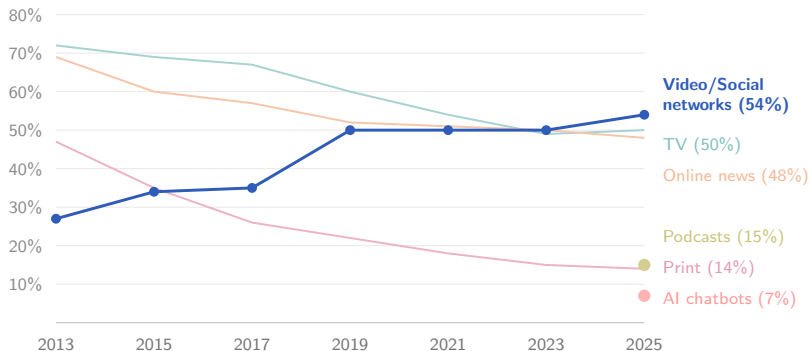


Computational Social Science with Images and Audio

Elliott Ash, Andrea Ciccarone, Philine Widmer

25 September 2026

Ciccarone (2026)



% of U.S. adults using each source for news in the past week. Source: Reuters Institute, Digital News Report 2025.

- ▶ How does partisan information/persuasion work in these multimodal environments?
- ▶ Focus so far almost exclusively on **text** partisanship
- ▶ **This paper:** images' properties make them particularly relevant in low-attention video settings
- ▶ Combination of **ML approach** and **Experiment**

Partisanship in Images and Text

Research task: measure partisan content separately in the images and language of political videos.

- ▶ **Labeled corpus:** 15,427 Democratic and Republican television ads from the Wesleyan Media Project, 2016–2020
- ▶ **Shared representation:** CLIP maps video frames and transcript segments into the same 512-dimensional space
- ▶ **Supervised measure:** train separate image and text classifiers to predict the same sponsor-party label
- ▶ **Interpretation:** decompose the dense embeddings into named concepts and test which concepts drive each prediction

Reusable workflow: **represent** → **predict** → **interpret**.

Representing Political Ads in CLIP Space



$$\downarrow$$

$$x_{\text{img}}^1 \in \mathbb{R}^{512}$$

"Dad helped build this business from scratch..."



$$\downarrow$$

$$x_{\text{img}}^t \in \mathbb{R}^{512}$$

"...protect Medicare and Social Security, especially with COVID..."



$$\downarrow$$

$$x_{\text{img}}^N \in \mathbb{R}^{512}$$

"My friends, my family, my community, that's why I'm running for Congress..."

$$\downarrow$$

$$x_{\text{txt}}^1 \in \mathbb{R}^{512}$$

$$\downarrow$$

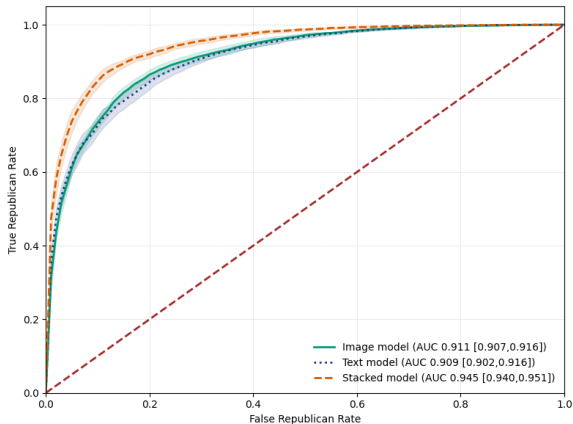
$$x_{\text{txt}}^t \in \mathbb{R}^{512}$$

$$\downarrow$$

$$x_{\text{txt}}^N \in \mathbb{R}^{512}$$

Raw videos $\xrightarrow[\text{pre-trained parameters}]{\text{CLIP}}$ $\{x_{\text{mod}}^1, \dots, x_{\text{mod}}^N\}$ $\xrightarrow{\text{aggregation}}$ $\mathbf{v}_{\text{mod}}$ $\xrightarrow[\text{trainable parameters}]{\text{neural network}}$ $P(\text{Rep} \mid \mathbf{v}_{\text{mod}})$

How Predictive Are Images and Text?



The same outcome and evaluation metric let us compare how much predictive information each modality carries.

Example of image-based partisanship predictions



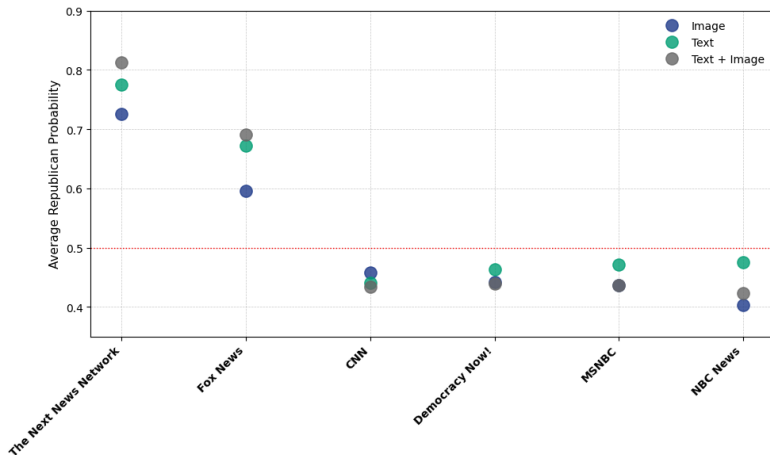
$P(\text{Rep}) = 0.01$



$P(\text{Rep}) = 0.99$

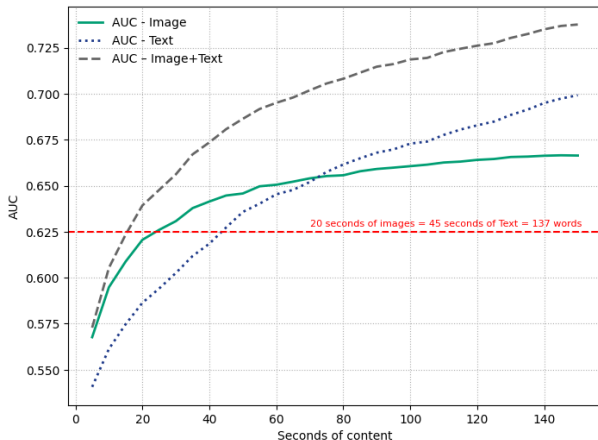
Notes: Two frames from immigration coverage on different channels, respectively MSNBC (left) and Fox News (right). Predicted probabilities are obtained from the image-only classifier trained on political ads.

Classifying Partisanship Across News Channels



The classifier was trained on political ads and then applied to news videos without retraining.

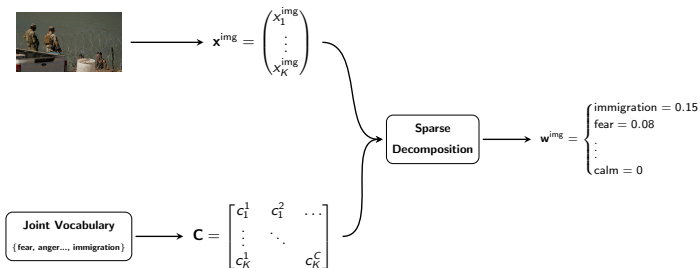
Images Convey Partisan Information More Quickly



At short exposure lengths, images carry more predictive signal than text.

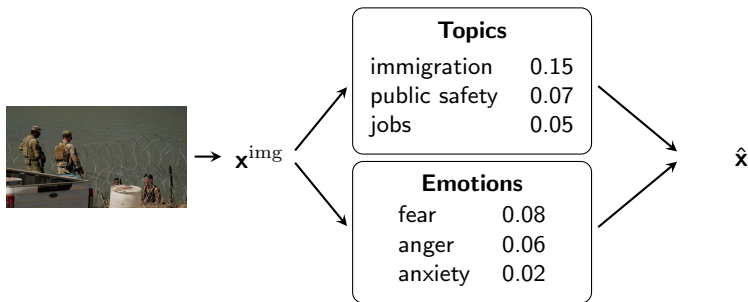
Opening the Black Box: Emotions vs. Topics

- Images convey partisan information quickly. Do images and text rely on **different concepts**?

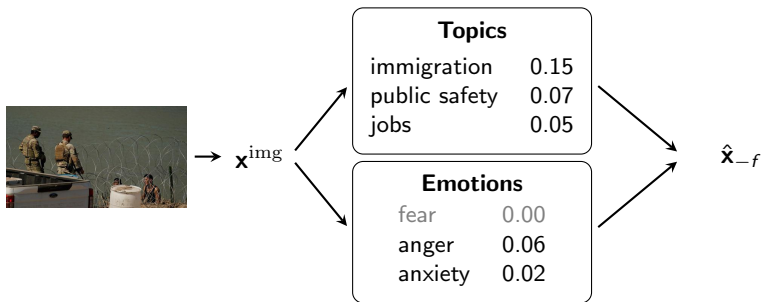


Intuition: represent videos as linear combinations of emotions and topics.

1. What happens to the representation if we remove a concept?



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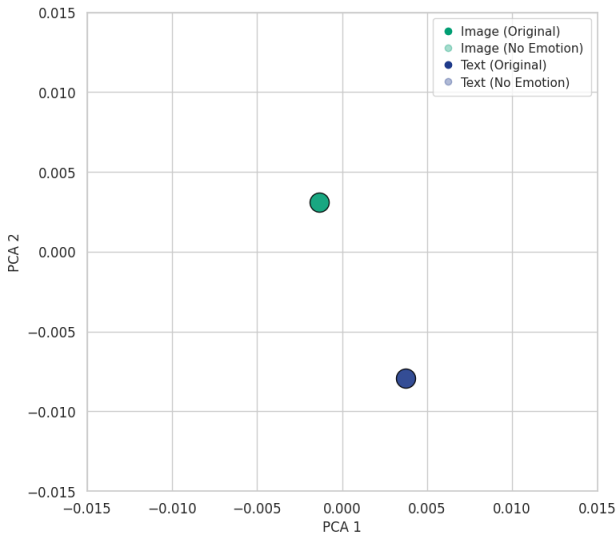


$$\hat{\mathbf{x}}_{-f} = C_T W_T + C_E W_{E,-f}$$

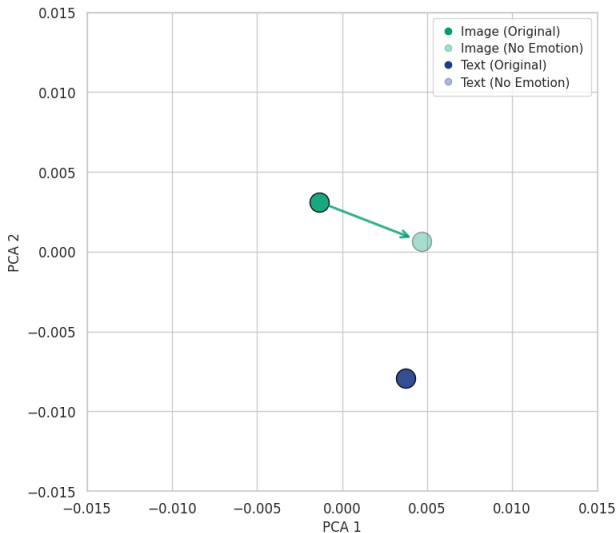
Set the weight of **fear** to zero.

→ Compare $\hat{\mathbf{x}}_{-f}$ with $\hat{\mathbf{x}}$.

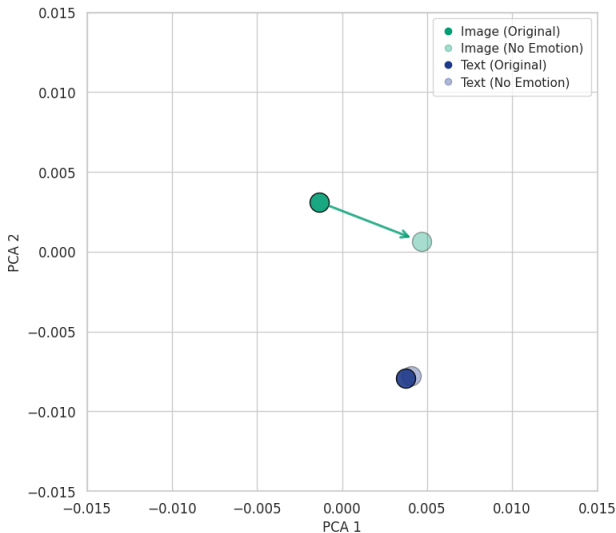
1. What happens if we remove a concept? Removing “Fear”



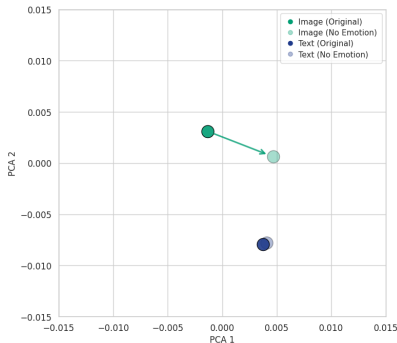
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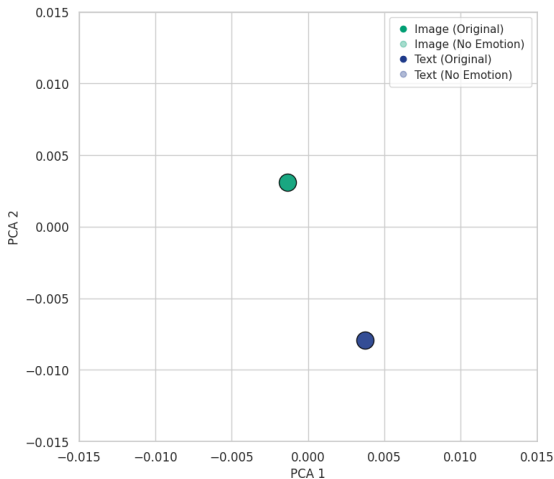


→ Removing a single **emotion** concept shifts images by $0.014 \cos \theta \simeq 9.6^\circ$ rotation.

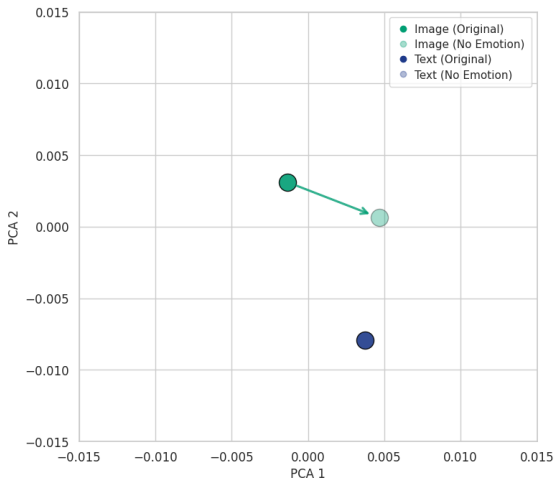
→ For text, the shift is $0.006 \simeq 6.3^\circ$.

→ For **topic** shifts, the difference is insignificant (13° vs. 12°).

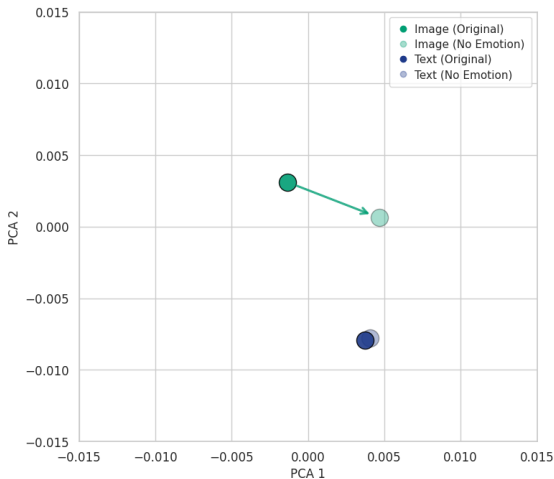
1. What happens if we remove a concept? Removing “Fear”



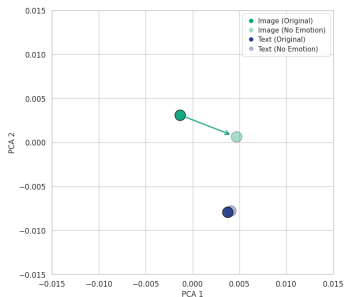
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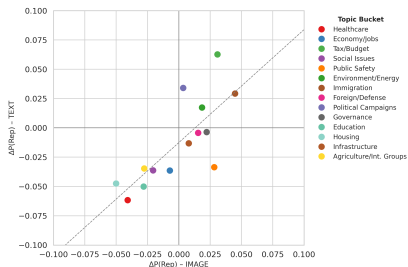
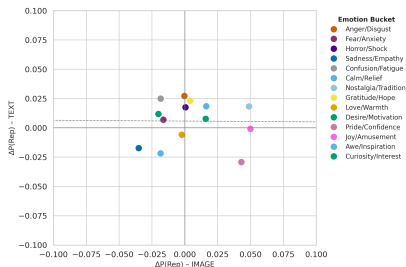


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Does Image Partisanship Rely More on Emotion?



→ Avg. absolute **emotions** CTE is 0.021 for image and 0.016 for text;
 $r = -0.01$

→ Avg. absolute **topics** CTE is 0.025 for image and 0.033 for text;
 $r = 0.76$

Experiment Design: Independently Varying Image and Text

Same event (TX SB4 hearing on immigration). AI voiceover standardizes delivery.

Republican Text

Democratic Text

Rep. Images



"illegal crossings... enforcement"



"families seeking asylum... rights"

Dem. Images



"illegal crossings... enforcement"



"families seeking asylum... rights"

Each cell $\times$ 2 lengths (30s / 120s) = 8 treatment arms

Treatment Arm: Republican Image × Republican Text

Republican Images (Fox News)



Republican Text (Fox News)

"The showdown between texas and the biden administration, now going one step higher on the legal ladder. appeals court set to hear oral arguments about a law that allows the lone star state to arrest and deport illegals only weeks after a lower court put that law on hold [...] the border patrol not doing their job so he wants to do his job by having the state enforce this law."

Main Specification

- ▶ **Sample:** $\sim 4,000$ participants (Prolific), $N = 3,430$ after restrictions
- ▶ **Design:** 8 treatment groups; pre-registered

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Consider this set of outcomes:

1. **Immigration topic attitudes** (eg. “Do you favor allowing U.S. military to assist immigration raids?”)
2. **Emotions:** ask to rate emotions (eg. fear, anger, disgust)
3. **Charity choice:** you can choose a prize of 25 dollars or you can donate a portion to pro-immigration charity

Text (Slowly) Affects Attitudes; Images (Quickly) Affect Emotions

$$Y_i = \beta_0 + \beta_1 \text{ImageR}_i + \beta_2 \text{TextR}_i + X_i^\top \gamma + \varepsilon_i$$

Text (Slowly) Affects Attitudes; Images (Quickly) Affect Emotions

$$Y_i = \beta_0 + \beta_1 \text{ImageR}_i + \beta_2 \text{TextR}_i + X_i^\top \gamma + \varepsilon_i$$

(a) Anti-Immigration			(b) Negative Emotions			(c) Charity Choice		
	Long	Short		Long	Short		Long	Short
Image (Rep)	-0.037 (0.029)	0.005 (0.028)	Image (Rep)	0.120** (0.048)	0.188*** (0.046)	Image (Rep)	0.062 (0.044)	-0.094** (0.044)
Text (Rep)	0.071** (0.029)	-0.013 (0.028)	Text (Rep)	0.001 (0.048)	0.031 (0.046)	Text (Rep)	0.023 (0.044)	0.023 (0.044)
Obs.	1748	1682	Obs.	1748	1682	Obs.	1748	1682

⇒ Text shifts attitudes only in long clips; images shift emotions and behavior immediately.

Conclusions

- ▶ Shift to short-form video changes the nature of political persuasion: filters out text effect on policy attitudes, leaving only the emotional effect driven by images
- ▶ Text-only measures are not fully equipped to study modern media environment
- ▶ Policies focusing only on text miss a central channel of modern political communication

In a low-attention video-first world, studying the visual channel is critical

Beraja, Yang, and Yuchtman (2023)

Henceforth referred to as BYY (2023)

JOURNAL ARTICLE

Data-intensive Innovation and the State: Evidence from AI Firms in China

Martin Beraja, David Y Yang ✉, Noam Yuchtman

The Review of Economic Studies, Volume 90, Issue 4, July 2023, Pages 1701–1723,

<https://doi.org/10.1093/restud/rdac056>

Published: 13 August 2022 **Article history** ▼

BYY (2023): Motivation

Henceforth referred to as BYY (2023)

- ▶ AI development requires massive amounts of data
- ▶ States often collect far more data than private sector
- ▶ Government data may exceed private data in:
 - ▶ Public security
 - ▶ Defense
 - ▶ Healthcare
 - ▶ Education
 - ▶ Basic science

BYY (2023): Research question & context

- ▶ **Key Question:** Does access to government data stimulate commercial AI innovation?
- ▶ **Context:** China's facial recognition AI industry
 - ▶ Leading AI sector at technological frontier
 - ▶ Firms need large training datasets (e.g., facial video data)
 - ▶ State collects extensive surveillance data
 - ▶ Firms gain access through government contracts

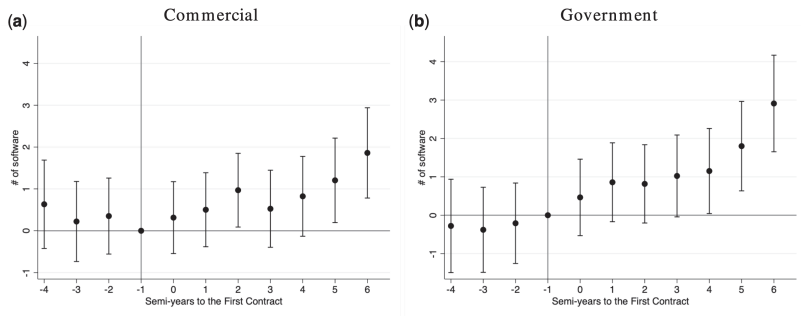
BYY (2023): Conceptual framework

- ▶ Two types of data:
 - ▶ Government data (accessed through contracts)
 - ▶ Private sector data
- ▶ Key mechanisms:
 - ▶ Direct effect: inputs shareable across uses (data, algorithms)
 - ▶ Indirect effect: inputs not shareable across uses (e.g., learning to manage large datasets)
- ▶ Potential trade-off: Resource allocation to government projects

BYJ (2023): Data & methodology

- ▶ Sample: 7,837 Chinese facial recognition AI firms
- ▶ Key measures:
 - ▶ Commercial & government software releases
 - ▶ Public security procurement contracts
 - ▶ Local surveillance capacity (cameras per capita)
- ▶ Research design:
 - ▶ Triple-differences approach
 - ▶ Compare effects of data-rich vs. data-scarce contracts

BYJ (2023): Results



Source: BYJ (2023). Differential cumulative software releases intended for commercial (left), for government uses (right), resulting from data-rich public security contracts, relative to data-scarce ones, controlling for firm and time period fixed effects. Data-rich contracts are defined as public security contracts in prefectures with above median surveillance capacity.

BYY (2023): Results ctd.

- ▶ Data-rich contracts lead to:
 - ▶ ~2 additional commercial software products over 3 years (cumulatively)
 - ▶ 20.2% increase relative to pre-contract level
- ▶ Concurrent increase in government software:
 - ▶ 2.9 additional products
 - ▶ 51.9% increase
- ▶ Mechanism: evidence of direct data sharing

BYY (2023): Broader implications

- ▶ Government data access can drive commercial AI innovation
- ▶ Applicable beyond facial recognition:
 - ▶ Geospatial data
 - ▶ Healthcare data
- ▶ Policy implications:
 - ▶ State procurement as innovation policy
 - ▶ Data provision strategies
 - ▶ International competitiveness (e.g., similar to discussion on privacy)
 - ▶ Important normative questions (e.g., civil liberties)
- ▶ Link to political econ debates on AI development (and innovation) in autocracies vs. democracies